Poisson 2026 Summer School

Diffeological Riemannian orbifolds

David Miyamoto

Queen's University

Diffeology

A *diffeological space* consists of a set X and an assignment to each $U \in \mathsf{Eucl}$ a set $\mathscr{D}_U \subseteq \underline{\mathsf{Set}}(U, X)$, whose elements are called *plots*, such that

- (Concreteness) All constant maps $U \to X$ are plots.
- (Pre-sheaf) If $f\colon U \to V$ is smooth, then $f^*\mathscr{D}_V \subseteq \mathscr{D}_U$.
- (Sheaf) If $f\colon U \to X$ is locally in a plot, then f is a plot.

A map $f\colon (X, \mathscr{D}^X) \to (Y, \mathscr{D}^Y)$ is *smooth* if $f_*\mathscr{D}^X \subseteq \mathscr{D}^Y$.

Dflg is a quasi-topos

All subsets, products, quotients, coproducts, and mapping spaces of diffeological spaces inherit natural diffeologies.

The D-topology

The *D-topology* of $(X, \mathscr{D})$ is the finest topology (most open sets) such that every plot is continuous.

Manifolds

Every $M \in \mathsf{Eucl}$ carries a natural diffeology:

$$\mathscr{D}_U^M := \{p\colon U \to M \mid p \text{ is } C^\infty\text{-smooth}\}.$$

The functor $y\colon \mathsf{Eucl} \to \mathsf{Dflg}$ is full, faithful, and injective on objects. We similarly get an embedding $\mathsf{Mfld} \to \mathsf{Dflg}$.

Diffeological orbifolds

Orbifolds, which generalize manifolds, are spaces that are locally isomorphic to quotients of $\mathbb{R}^n$ by finite group actions:

A diffeological space X is an *orbifold* if for each $x \in X$, there is:

- a D-open neighbourhood A of x;
- some finite group action $\Gamma \circlearrowright \mathbb{R}^n$,
- a D-open neighbourhood V of $\mathbb{R}^n/\Gamma$,
- and a diffeomorphism $\varphi\colon A \to V$.

Diffeological orbifolds, and smooth maps between them, coincide with the usual notions found in the literature.

References

[dF19] M. del Hoyo and R. L. Fernandes. "Riemannian metrics on differentiable stacks". In: *Math. Z.* 292.1-2 (2019), pp. 103–132.

[KM25] Y. Karshon and D. Miyamoto. "Quasifold groupoids and diffeological quasifolds". In: *Transform. Groups* 30.3 (2025), pp. 1333–1367.

[KSS25] K. Kuribayashi, K. Sakai, and Y. Shiobara. "Towards Riemannian diffeology". In: *Proc. R. Soc. Edinb., Sect. A, Math.* (2025), pp. 1–30.

[Miy26] D. Miyamoto. *Diffeological Riemannian orbifolds*. 2026. arXiv: 2607.08939 [math.DG].

Metrics in Dflg I

Covariant tensors

Let $\hat{\mathcal{T}}^k\colon \mathsf{Eucl}^{\mathrm{op}} \to \mathsf{Dflg}$ be the functor that sends U to the space of covariant k-tensors,

$$\hat{\mathcal{T}}^k U := \Gamma\left(\bigotimes^k T^*U\right).$$

We set $\mathcal{T}^k\colon \mathsf{Dflg} \to \mathsf{Dflg}$ to be the right Kan extension of $\hat{\mathcal{T}}^k$ along y. An element of $\mathcal{T}^k X$ is a *covariant k-tensor*.

Concretely, a k-tensor on X is an assignment to each plot $p\colon U \to X$ a covariant k-tensor $\alpha_p \in \hat{\mathcal{T}}^k U$, such that if $f\colon V \to U$ is smooth, then $\alpha_{p\circ f} = f^*\alpha_p$.

Definition

A *weak Riemannian metric* on X is a 2-tensor g such that g_p is positive and symmetric for all plots p.

Metrics in Dflg II

Tangent spaces

Let $\hat{T}\colon \mathsf{Eucl} \to \mathsf{Dflg}$ be the functor that sends $U \subseteq \mathbb{R}^n$ to its tangent bundle,

$$\hat{T}U := y(U \times \mathbb{R}^n).$$

We set $T_k\colon \mathsf{Dflg} \to \mathsf{Dflg}$ to be the left Kan extension of the k-fibred tangent functor $\hat{T}_k = \hat{T} \times_\pi \cdots \times_\pi \hat{T}$.

Concretely, an element of $T_k X$ is represented by a plot $p\colon U \to X$ and a tuple $(u, v_1, \ldots, v_k)$ in $U \times (\mathbb{R}^n)^k$.

Definition [KSS25]

A *weak Riemannian metric* on X is a smooth map $g\colon T_2X \to \mathbb{R}$ such that $g_p := g \circ T_2 p$ is a positive and symmetric covariant 2-tensor on U for all plots p.

Warning: $T_2 \ncong T \times_\pi T$.

Metrics on Lie groupoids

A Riemannian metric η on E is *transverse with respect to* a surjective submersion $p\colon E \to B$ (or simply *p-transverse*) if the maps

$$T_e(E_b)^\perp \to T_{e'}(E_b)^\perp$$

are all isometries.

Let $\mathcal{G} \rightrightarrows M$ be a Lie groupoid, with underlying (singular) foliation $\mathcal{F}$.

- A *2-metric* is a metric on $\mathcal{G}\ {}_s\!\times_t \mathcal{G}$ that is m-transverse and S_3-invariant.
- A *1-metric* is a metric on $\mathcal{G}$ that is s-transverse and S_2-invariant.
- A *0-metric* is a metric η on M such that:
  - if $I\colon T\mathcal{F}^\perp \to N\mathcal{F}$ denotes the smooth bijection induced by η,
  - and η^N denotes the bundle-metric on $N\mathcal{F}$ induced by η and I,
  - then η^N is invariant under the $\mathcal{G}$-action on $N\mathcal{F}$, where

$$g \cdot [Ts(\xi)] := [Tt(\xi)],\ \xi \in T_g\mathcal{G}.$$

Orbifold groupoids

An *orbifold groupoid* is an étale and proper Lie groupoid. A 0-metric on an étale groupoid is one that is invariant under the effect pseudogroup.

Metrics on stacks [dF19]

A *2-metric* on a stack is an equivalence class of 2-metrics (equivalent ones induce the same 0-metric).

The main result

Theorem [Miy26]

a) A 0-metric η on $\mathcal{G}$ induces g on $M/\mathcal{G}$ if and only if the following commutes (λ is natural):

$$\begin{array}{ccc} N_2\mathcal{F} & & \\ \downarrow{\scriptstyle \lambda} & \searrow^{\eta^N} & \\ T_2(M/\mathcal{G}) & \xrightarrow{g} & \mathbb{R} \end{array}$$

b) If $\mathcal{G}$ is not regular, then no 0-metric on $\mathcal{G}$ can induce one on $M/\mathcal{G}$.

c) If $\mathcal{G}$ is regular and proper, then a 2-metric on $\mathcal{G}$ always induces a weak metric on the orbit space.

d) If Γ is countable and acts freely (not necessarily properly) on M, then a Γ-invariant metric always induces a weak metric on M/Γ.

Write $F = T_2$. Here are some of the key steps.

- η^N is continuous (and smooth) if and only if $\mathcal{F}$ is regular. Use that η^N_x is non-degenerate.
- η^N always induces a map $FM/F\mathcal{G} \to \mathbb{R}$.
- The natural map $FM/F\mathcal{G} \to F(M/\mathcal{G})$ is an isomorphism if:

$$\text{whenever } (p,q)\colon U \to M \times_{M/\mathcal{G}} M, \text{ then also } (Fp, Fq)\colon FU \to FM \times_{FM/F\mathcal{G}} FM. \tag{1}$$

- Condition (1) holds if (locally) $(p,q) = (t,s) \circ g$ for some map $g\colon U \dashrightarrow \mathcal{G}$ (true for d)).
- For proper $\mathcal{G}$, we have the above factorization on each stratum; then (1) holds on an open dense subset of U, hence everywhere (by regularity).

Diffeology also gives a new interpretation of Myers-Steenrod for orbifolds. Next: does a Myers-Steenrod theorem hold for quotients as in d)?

From groupoids to diffeology

There is a functor Q from the bicategory Grpd of (Lie groupoids, principal bibundles, isomorphisms of principal bibundles) to Dflg (trivial 2-arrows), defined by

$$Q(\mathcal{G} \rightrightarrows M) := M/\mathcal{G}.$$

The functor Q induces an equivalence

$$(\text{orbifold grpds, Morita bibundles}) \xrightarrow{\cong} (\text{dflg orbifolds, diffeomorphisms})$$

An aside

We can replace "orbifold" with "quasifold," and "Morita bibundles/diffeos" with "submersions." [KM25]

Induced metrics

Consider two maps:

- The pullback $\pi^*\colon \mathcal{T}^k(M/\mathcal{G}) \to \mathcal{T}^k M$.
- The dual of the quotient $TM \to N\mathcal{F}$, pushing forward sections: $j\colon \Gamma\left(\bigotimes^k N^*\mathcal{F}\right) \to \mathcal{T}^k M$.

We say a 0-metric on $\mathcal{G}$ *induces* a weak Riemannian metric g on $M/\mathcal{G}$ if $\pi^* g = j(\eta^N)$.

LaTeX TikZposter